Paleomagnetism of Carboniferous diabase dykes from Gaspé, Quebec

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The reported paleomagnetic study was carried out on 53 oriented samples (156 specimens) at 10 sites in the eastern Gaspé Peninsula. All sampled geological units are composed of subvertical diabase dykes that cut the sedimentary formations of late Early Devonian – early Middle Devonian age. The radiogenic (whole-rock K/Ar) age of these and similar dykes ranges from Late Devonian to early Middle Carboniferous (mean whole rock K/Ar age = 310 Ma). Two components were isolated. The most common component, A ($D = 152°$, $I = +41°$), is normal, whereas the second component, B ($D = 315°$, $I = -44°$), is reverse. Both components may be considered as a single one; alternatively, component A may be considered slightly older than component B. Unblocking temperatures, median destructive fields, relative frequency of occurrence of the components, and various degrees of alteration are arguments favouring the first or the second situation. The case of a single component is more probable, representing an Early Carboniferous time of acquisition of remanence (thermoremanent magnetization (TRM) probably corresponding to the time of intrusion and resetting of surrounding sedimentary rocks). The corresponding paleopole position in 148°E, 11°N. This pole position is located some 25° to the southeast of the cluster of Carboniferous poles (mainly Middle to Late Carboniferous) for the North American craton. The significance of this discrepancy is not well understood, but it is worth noting that very recent paleomagnetic results from western Newfoundland and central New Brunswick are located closer to the paleomagnetic results of this study. The age and direction of the dyke system suggest a short period of extension following the compressive pulse of the Acadian orogeny and preceding the Alleghenian orogeny. For these reasons, the dyke system is unrelated to the initial opening of the present Atlantic Ocean.

Une étude paléomagnétique a été complétée sur 53 échantillons orientés (156 spécimens) prélevés sur 10 sites dans la partie orientale de la péninsule de la Gaspésie. Toutes les unités géologiques échantillonnées sont formées de dykes de diabase subverticaux qui recoupent les formations sédimentaires d'âge du Dévonien supérieur tardif – début du Dévonien moyen. L'âge radiogénique (K/Ar sur roche totale) de ces dykes, et autres semblables, varie du Dévonien supérieur au début du Carbonifère moyen (310 Ma = moyenne des âges K/Ar sur roche totale). Deux composantes ont été isolées. La composante la plus fréquente, A ($D = 152°$, $I = +41°$), est normale, tandis que la deuxième composante, B ($D = 315°$, $I = -44°$) est inversée. Les deux composantes peuvent être interprétées comme une seule, mais on peut alternativement considérer la possibilité que la composante A soit légèrement plus ancienne que la composante B. Les températures de déblocage, les champs destructeurs moyens, la fréquence relative d'apparition des composantes et les degrés différents d'altération favorisent aussi bien la première que la seconde interpretation. Cependant, le cas d'une composante unique semble plus probable, et la période de l'acquisition de la rémanence correspondrait au Carbonifère inférieur (l'aimantation rémanente thermique concorde probablement avec le temps de l'intrusion et de la remise à zéro des roches sédimentaires encaissantes). La position du paléopôle correspondant est 148°E, 11°N. Cette position du pôle est éloignée de quelque 25° au sud-est du groupe des pôles carbonifères (principalement Carbonifère moyen à supérieur) pour le craton de l'Amérique du Nord. La raison de cet écartement n'est pas bien connue, mais il est intéressant de noter que les résultats des études paléomagnétiques les plus récentes pour la partie occidentale de Terre-Neuve et pour la partie centrale du Nouveau-Brunswick fournissent des pôles qui se rapprochent de ceux de notre étude. L'âge et la direction du système de dykes suggèrent l'existence d'un stade de distension de courte durée après la poussée en compression due à l'orogenèse de l'Acadien mais précédant l'orogenèse de l'Alléghénien. Par ces raisons, ces dykes ne pas associés au début de l'ouverture de l'océan Atlantique actuel.

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Introduction

Despite extensive paleomagnetic studies over the last 30 years, the entire Paleozoic apparent polar wander path (APW) for North America is still ill defined: the best data set is for the Carboniferous–Permian time interval. A great many pole positions from Devonian rock units lie within 15–20° of the Permo-Carboniferous poles. This implies that either North America moved very little relative to the paleopoles in this time interval or Permo-Carboniferous overprints dominate (Creer 1968). As a great number of paleomagnetic studies originate from the Appalachians, enough data became available to compare them with those from cratonic North America. The apparent latitudinal discrepancy between the two data sets led to the hypothesis of sinistral strike-slip motion (Kent and Opdyke 1978) and (or) rotation (Spariosu and Kent 1983) of specific Appalachian terranes. Recent studies (Irving and Strong 1984, 1985; Seguin *et al.* 1985; Murthy 1985) have demonstrated that this first hypothesis is very unlikely, but the second may still be applicable occasionally. Because the ideas of relative motion have been favoured until recently, the possibility of a widespread overprinting of cratonic North American rocks was set aside. This led to a false interpretation of the late Paleozoic history of eastern North America (Irving and Strong 1985).

For the very Late Devonian and Carboniferous periods, many paleomagnetic data are available in North America. However, the large scatter of paleopoles and paleolatitudes complicates the interpretation of all these paleomagnetic results. Many attempts were made to obtain reliable Devonian and Carboniferous paleopole positions and APWP's for cratonic North American (Irving 1979; Van der Voo 1981; Roy *et al.* 1983; Roy and Morris 1983; Irving and Strong 1984; Seguin and Gahé 1985; Seguin 1985). For various reasons (e.g., insufficient data average, magnetic overprints, inaccurate dating), the Devonian APWP and to a lesser extent the Carboniferous APWP for the North American craton are not

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well defined. Consequently, one interpretation implies little polar motion and the other, great polar motion. One of the goals of this study is the addition of new information to the existing data base and if possible the obtainment of a more reliable definition of the late Paleozoic APWP.

Another objective of the study is the determination of the nature and origin of these diabase dykes. It is important to establish a temporal relationship of the intrusions with respect to the sedimentary host rocks. It is then worth comparing the magnetic signature of the present group of dykes with Jurassic and Triassic dyke systems from the trans-New Brunswick lineament, Grand Manan Island, North Mountain and Shelburne in Nova Scotia, the Trans-Avalon lineament in Newfoundland (Seguin *et al.* 1981), and Anticosti Island in Quebec (Seguin and Petryk 1984). The dyke systems are oriented northeast, whereas the system studied has an east–west orientation. The Late Triassic – Jurassic igneous activity is related to the opening of the present Atlantic Ocean and the formation of sedimentary basin on the continental margin offshore Nova Scotia, Newfoundland, and Labrador. Given the important hydrocarbon potential of these basins, the study of the dyke systems is important for understanding regional stresses affecting the formation of the basins.

Another interesting aspect of dyke investigation deals with the seismo-tectonic regime of New Brunswick and eastern Quebec. Some dyke systems coincide with the opening time of the Atlantic Ocean. Which type and magnitude of tectonic regime affected the North American plate throughout geological ages? There is evidence (e.g., basic and ultramafic dykes) that the eastern North American underwent plate extension in Late Precambrian time, as indicated by the St. Lawrence rift (Kumarapeli and Saull 1966), before or during the formation of the proto-Atlantic Ocean, two or three subsequent phases of deformation (in compression) in early and late Paleozoic time, then extension in Triassic–Jurassic time and recent compression, based on fault-plane solutions obtained from earthquakes in the Charlevoix region of Quebec and the Miramichi area in New Brunswick (Seguin 1982; Stevens 1980). As the investigated dyke system is located between the two localities mentioned above, a partial explanation of this situation may be obtained.

Geological setting and sampling sites

The area investigated is located in the Gaspé – Connecticut Valley synclinorium of the Quebec Appalachians. Geological descriptions of this region may be obtained in McGerrigle (1950), Rust (1976), Bourque (1977), Béland (1978), Sikander (1978), Brisebois (1979), and Bourque and Lachambre (1980).

The diabase dykes are composed of well-crystallized plagioclase and pyroxene with a subophitic texture. The diabase at Tar Point (site 9) is vesicular, and the vesicles were locally filled with bitumen after cooling of the dyke (Fig. 1). The hydrocarbons in the host rocks migrated upward into the cavities within the dyke. The dykes are subvertical and cut the Early Devonian (Gedinnian; 390 Ma) aged Indian Point Formation as well as the Early to early Middle Devonian (late Emsian; 370 Ma) aged York River and Battery Point formations. The Early Devonian sedimentary units were deformed during the Late Devonian sedimentary Acadian orogeny ($\simeq$350 Ma). The diabase dykes definitely postdate this orogenic event, since they cut through the folded rock units. Five sampling sites (Fig. 1) are located on the north side of Gaspé Bay, while site 6 is situated on the south side of Gaspé Bay. The first five sites are roughly oriented along an east–west direction. Site 7 is located on Gravelle Creek, site 8 south of Seal Cove River, and site 10 on La Chesnaye Creek. Sites 7, 8, 9, and 10 all line up in an east–west direction. The dykes to the north and south of Gaspé Bay appear to be parts of two long dykes cut by local and minor orthogonal faults. The age of the dykes is not known with great accuracy, but given their intrusive relationship with the sedimentary host rocks and their radiogenic ages, the age bracket can be reasonably constrained. Based on geological inference, McGerrigle (1950) presumed the dykes were of Late Devonian or Carboniferous age. The dykes dated by the K/Ar radiogenic method and representative of the nine dykes around and near Gaspé Bay mapped by McGerrigle (1950) usually range from 290 to 320 Ma (Wanless *et al.* 1966, 1973). These dates include that of one of the dykes in the present study. The dykes are thus most probably of Carboniferous age, and there is little evidence indicating that their intrusion age can be extended to Late Devonian time. Still, it is worth noting (1) excess argon is mentioned in Wanless *et al.* (1973) in a basalt from Grande Rivière, (2) older ages ranging from 317 to 357 Ma were obtained, (3) older ages up to 70 Ma were observed in more altered sills, (4) excess argon is not homogeneously distributed throughout the sills, and (5) the basalts radiogenically dated are correlated with the diabase sills.

Measurements and natural remanent magnetization (NRM) results

The equipment used for this study consists of (1) Schonstedt spinner magnetometer (DSM-1) for remanence measurements, (2) Cenco MS-3B bridge for susceptibility measurements, (3) Schonstedt thermal demagnetizer (TSD-1), and (4) Schonstedt alternating-field demagnetizer (GSD-1). NRM directions of the dykes (Fig. 2). are most frequently oriented in the northwest quadrant, and slightly over 30% of them coincide with the present Earth's field (PEF). An old component of magnetization is preserved, since an antiparallel direction is occasionally present in the southeast quadrant. The NRM intensities (J_n) range from 10^{-2} to 10^2 A m^{-1}, with values of the Koenigsberger ratio ($Q_n = J_n/kH$) between 0.3 and 40 (mean = 4).

Demagnetizations

All Carboniferous dykes were demagnetized thermally (20–600°C) and with alternating field (AF) (up to 100 mT). Figure 3 shows variations in orientation and intensity of magnetization for samples originating from four different sites after AF demagnetization. Sites 3 and 4 are located to the north of Gaspé Bay; site 7, to the south.

Normal polarities are found on sites situated to the north of Gaspé Bay, while site 7 located to the south has a reverse polarity. The west-northwest directions are characterized by a median destructive field (MDF) of 15–80 mT and unblocking temperature (T_{UB}) of 450–570°C. Specimen G1-5BA (Fig. 3) shows an example of the specimens containing a low-coercivity PEF component after AF demagnetization. For the south-southeast directions, MDF values range from 10 to 30 mT and the T_{UB} values range between 40 and 525°C, which are slightly lower coercivities and T_{UB}'s than for the west-northwest directions of magnetization. A similar although reverse situation was observed in Carboniferous volcanics from New Brunswick (Seguin *et al.* 1985). In general, AF and thermally demagnetized results are quite comparable. Figure 4 shows examples

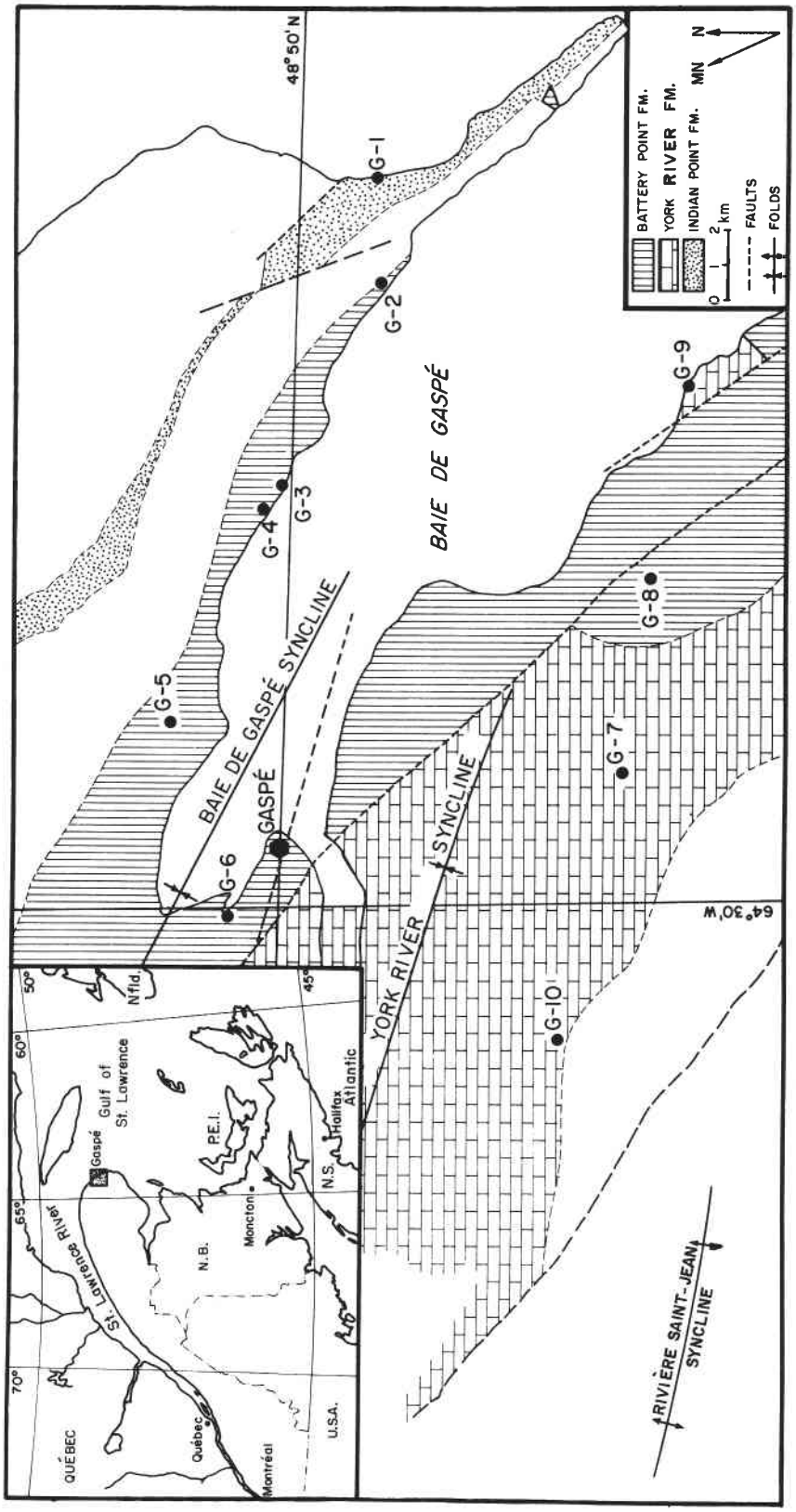


FIG. 1. Location map and simplified geological map of the eastern Gaspé area, province of Quebec. The sampling localities of the diabase dykes are indicated and numbered. MN, magnetic north; N, true north. Map modified after Brisebois (1979).

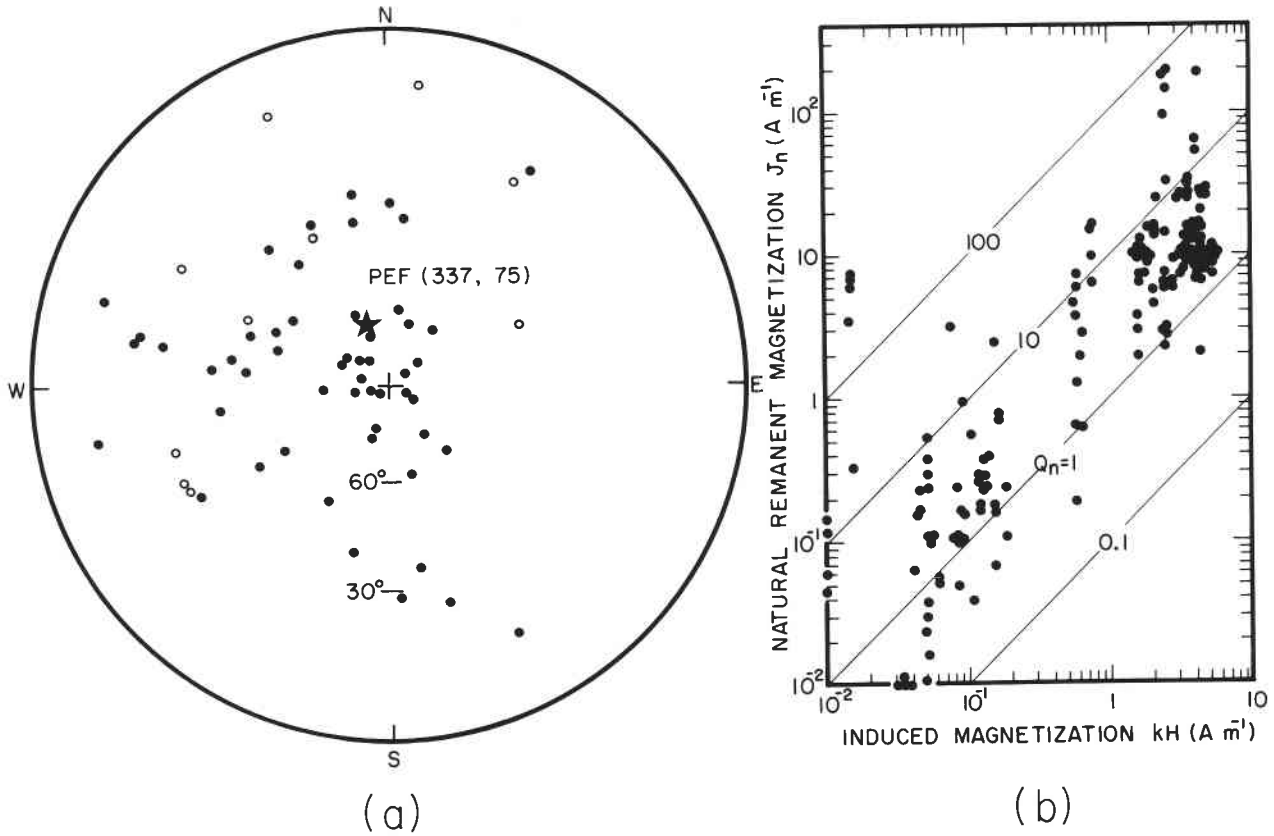


FIG. 2. (*a*) Individual sample NRM directions plotted on Wulff net for the diabase dykes. Solid (open) circles represent projections on lower (upper) hemisphere. The star represents the present Earth's field (PEF) at the sampling region. (*b*) Plots of corresponding intensity of NRM (J_n) against the induced magnetization (kH), with resulting slope lines indicating the values of Koenigsberger ratio, Q_n.

of thermally demagnetized specimens for both normal and reverse polarities. In specimen G3-5E (Fig. 4), a low-coercivity PEF component is present in the 20−350°C range. The memory carrier is located in both magnetite and titanomagnetite, as verified from isothermal remanent magnetization (IRM) experiments, coercivity spectra, and reflected-light microscopic observations.

Mean directions of magnetization

Figure 5 and Table 1 show the mean-sample directions observed in the diabase dykes. A comparison of all the directions of magnetization before (Fig. 2) and after cleaning (Fig. 5) indicates clearly that the demagnetization techniques used were efficient. Two groups of magnetization are observed: one of normal polarity (component A) has a west-northwest direction, and another of reverse polarity plots in the south-southeast quadrant (Fig. 5). The south-southeast directions (152°, +41°) and the north-northwest directions (315°, −44°) are within the confidence limits (8 and 10°, respectively,) with respect to the mean (324°, −43°). Therefore, on statistical grounds, components A and B are indistinguishable.

Still, there is some evidence to support a difference between components A and B. (*i*) The altered top parts of the cored samples often carry a PEF component that, once removed, gives way to the B component but never to the A component. (*ii*) The sampling sites located to the south of Gaspé Bay (e.g., G-7, G-8, G-9, and G-10) are more altered than those located to the north (e.g., G-1, G-2, G-3, and G-4). The regional geologic setting is the same for the two groups of dykes, but a study of their petrography shows marked differences in the degree of alteration of the dykes. Even though patchy alteration is present in the northern group of dykes, phenocrysts of augite, labradorite laths, and olivine (partly altered to iddingsite) are most commonly observed in the fresh dykes. In the southern group of dykes, augite is completely altered to amphiboles, the feldspars are highly sericitized, and the olivine is completely altered to chlorite, limonite, and possibly serpentine. Component A detected in sites G-2 and G-5 is present in the more altered samples as well. (*iii*) The iron oxides are often alteration products and usually coarser grained in the southern group of dykes. This situation may explain the lower T_{UB} and MDF of component A. (*iv*) The northern and southern groups of dykes cut the same formations and are not affected by a specific tectonic feature. These considerations suggest that component A may be older than component B. There is, of course, the possibility that the two sets of dykes are of different ages. This can be ruled out more easily because component A is also found at sites G-2, G-5, and G-6 from the northern set of dykes.

Age of acquisition of the components

The thickness of the dykes varies from 3 to 8 m, and the farthest sampling distance in the host rock outside the limit of a dyke varies from 6 to 20 m depending on the thickness of the dyke investigated. The paleomagnetic baked-contact test (PBCT) is an important tool to bracket the age of magnetization of the two components.

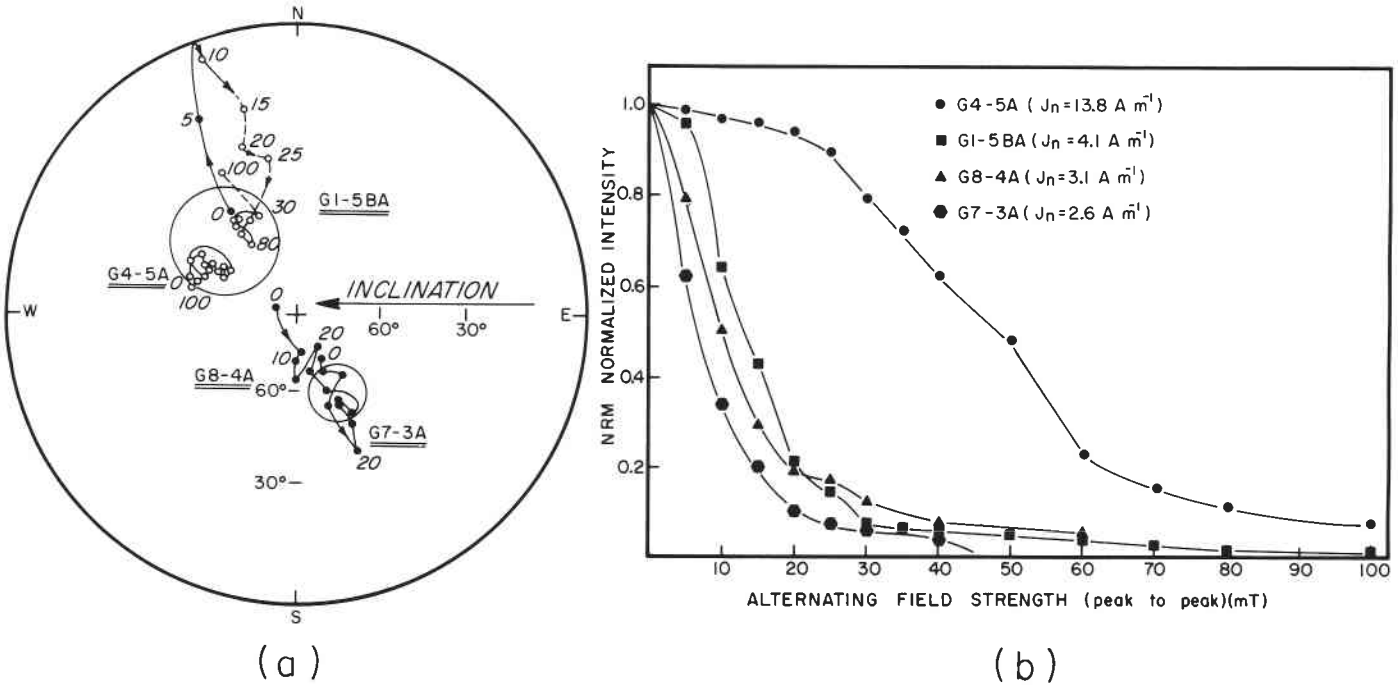


FIG. 3. (*a*) Wulff net projections of directions of magnetization of four specimens (two for component A and two for component B) from four different sampling sites after stepwise AF treatment. The numbers next to open and soild circles represent the demagnetization steps in milliteslas as indicated. (*b*) The respective normalized AF decay curves.

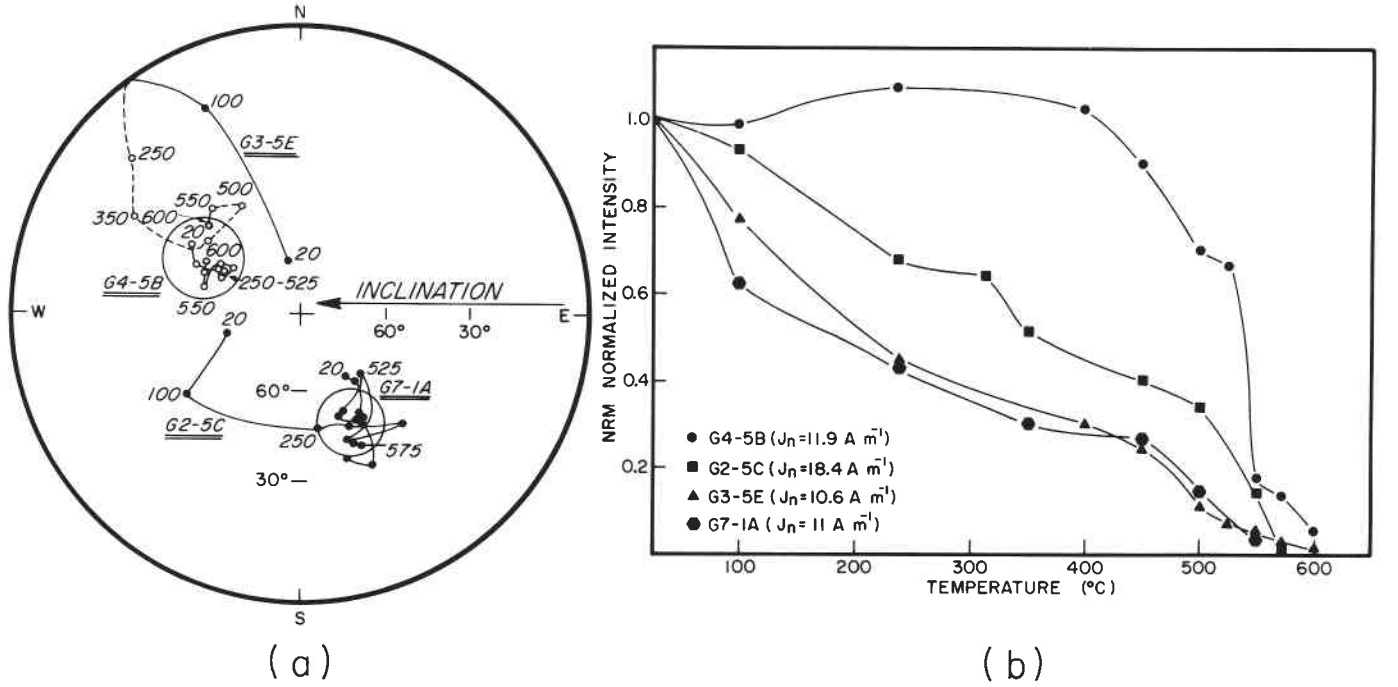


FIG. 4. (*a*) Wulff net projections of directions of magnetization of four specimens (two for component A and two for component B) from four different sampling sites after stepwise thermal treatment. The numbers next to open and solid circles represent the demagnetization steps in degrees Celsius as indicated. (*b*) The respective normalized AF decay curves.

Figure 6 shows some results of the baked-contact test at four diabase dyke sites (G-4, G-7, G-9, and G-10). At sites G-4, G-10, and possibly G-7, *it appears that the mean direction of component A or B is different from that of the mean component for the Lower Devonian sedimentary rocks.* Figure 7 shows a stereonet plot of the site-mean vector direction of the *in situ* component with its corresponding α_{95} cone of confidence (Fisher statistics) inside the dyke and outside (in the sedimentary rocks). This analysis shows that the PBCT is definitely positive at sites G-2, G-4, and possibly G-10. It is marginal at sites G-7 and G-9. Thus, at 30 to possibly 50% of the sites, a positive PBCT is observed, and it can be concluded

TABLE 1. Gaspé dykes, site-mean remanence directions

Site No.	N (n)	D (°)	I (°)	α_{95} (°)	k	P	
G-1	5 (17)	303	−45	7.7	99.7	R	
G-2	4 (17)	337	−31	12.7	53.2	R	
	1 (4)	164	37			N	
G-3	5 (23)	306	−47	3.9	370	R	
G-4	5 (17)	304	−48	10.1	57.6	R	
G-5	4 (11)	307	−41	12.7	37.3	R	
	3 (6)	159	41	37	15.7	N	
G-6	5 (20)	326	−48	24.5	11.2	R	
	2 (3)	135	42			N	
G-7	5 (17)	152	51	18.3	13.6	N	
G-8	5 (13)	147	49	18.5	17.7	N	
G-9	6 (13)	153	30	11.0	37.9	N	
G-10	5 (14)	153	33	18.4	18.5	N	
Mean	6 (105)	315	−44	10.3	43.3	R	B
Paleopole:		155°E,	6°N	dp=8.1°,	dm=12.9°		
	7 (70)	152	41	7.8	60.5	N	A
Paleopole:		142°E,	14°N	dp=5.7°,	dm=9.5°		
	13 (175)	324	−43	6.6	39.9	R+N	A+B
Paleopole:		148°E,	10°N				

NOTES: N is the number of samples; n is the number of specimens; D is the declination; I is the inclination; α_{95} is the semi-axis of 95% confidence level; k is Fisher's precision parameter; and P is the polarity—N is normal and R is reverse.

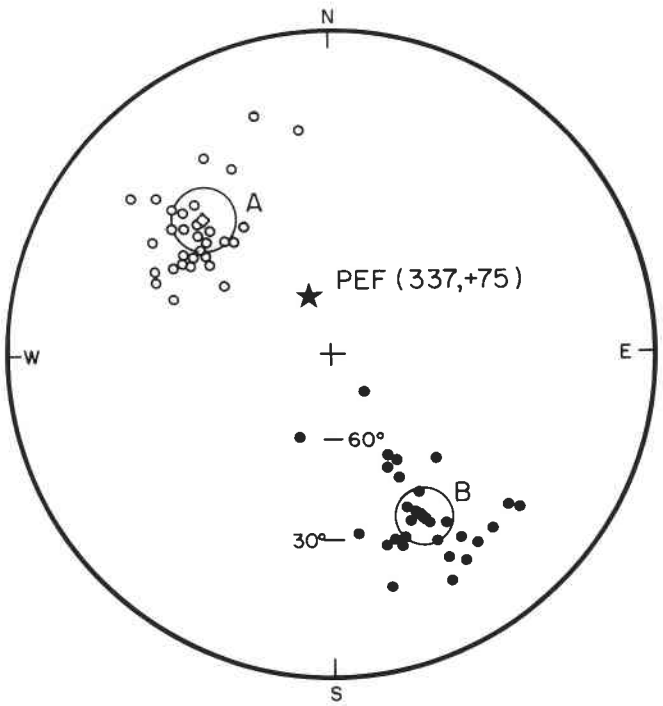


FIG. 5. Wulff net projections of sample-mean directions of magnetization obtained for the diabase dykes after AF and thermal treatments: for the two components (A and B). Solid (open) circles represent projections on lower (upper) hemisphere. Diamonds represent the site-mean direction of the two components along with the error circles (α_{95}). For comparison, the present Earth's field (PEF) is shown on the stereonet.

that the contact test is more frequently negative. At site G-10 (Fig. 6), the mean direction of magnetization of the dyke is almost *antiparallel* to that of the host rock. This observation may be interpreted as a younger component overprinted during a *polarity reversal interval*, in which case the PBCT is negative. Dual polarity is typical of Carboniferous times immediately preceding the onset of the Kiaman Magnetic Interval (Roy and Morris 1983); the 310 Ma dykes are just below this transition. Alternatively, the parallelism of the reversed component in the dyke and in the sediments near the contact may favour a positive PBCT. Consequently, this specific case is ambiguous.

What does the PBCT tell us about the relative age of the components? A positive PBCT is a strong indication, even though not a definite one, that the magnetization of the intrusive is primary or close to it. The mean direction of sites G-2 and G-4, which passed the PBCT, is 321°, −40° and that of sites G-7 and G-8 is 149°, 50° (329°, −50°, N). It becomes clear that the directions of components A and B are not significantly different (mean 325°, −45°, N). If the mean directions of sites G-9 and G-10 are added to those of G-7 and G-8, the new direction of component A becomes 149°, 41° (329°, −41°,N). In the last situation, the mean directions of components A and B are almost identical. The mean of the directions of components A and B (325°, −41°) is comparable with that obtained for all the sites (324°, −43°, Table 1). Thus, *the results of the PBCT strongly suggest that components A and B constitute only one single component*, which occurred in a very short time interval (≤1 Ma), especially on site G-10.

Discussion and conclusion

The paleopole obtained for component B is 155°E, 6°N, (dp = 8.1°, dm = 12.9°) and for component A (normal polarity), 142°E, 14°N (dp = 5.7°, dm = 9.5°). The corresponding paleolatitudes are 26°S and 23°S. The paleopole of component B is characterized by a latitude located south (Fig. 8) of most Early−Middle Carboniferous poles of North America. Various possible explanations can be given for this situation. (*i*) The crystallization of the dykes spans a very short period of time, and the secular variations are not averaged; only a virtual geomagnetic pole is recorded. (*ii*) Most of the

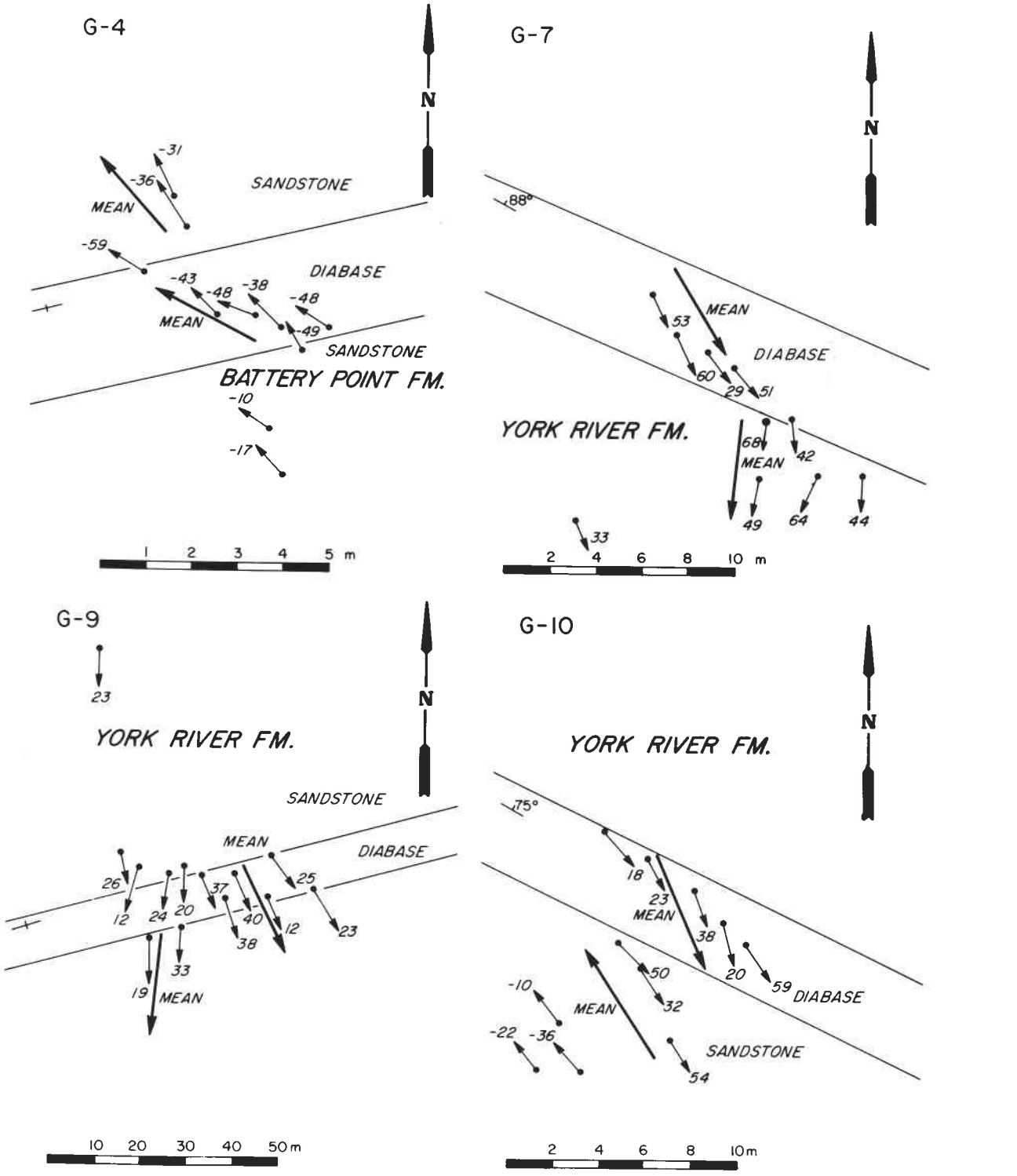


FIG. 6. Representative examples of the paleomagnetic baked-contact test (PBCT) taken from four sites (G-4, G-7, G-9, and G-10) as indicated. Each dot represents a sampling location. The small arrows show the direction of magnetization (declination); the numbers give the inclination (degrees) with respect to the horizontal plane. The sample-mean directions for the sedimentary host rocks and the diabase dykes are represented by the long arrows. At some sites, the PBCT could be performed on only one side of the diabase dykes.

poles clustered around 130°E, 35°N (Fig. 8) are obtained from continental sedimentary rock units, while the poles resulting from this study and the New Brunswick volcanics (NB) are located farther southeast. It is thus possible that detrital remanence and thermoremanence did not record the same field orientation during the same time interval. Still, the Carboniferous Deer Lake Group (DL) and Spout Falls formation (SP) from southeastern and western Newfoundland, respectively, are both sedimentary units, and their paleopoles are also situated southeast of the main cluster of poles. A post-Early Carboniferous rotation of Newfoundland could explain this anomaly, but Deutsch and Rao (1977) did not find any evidence for such

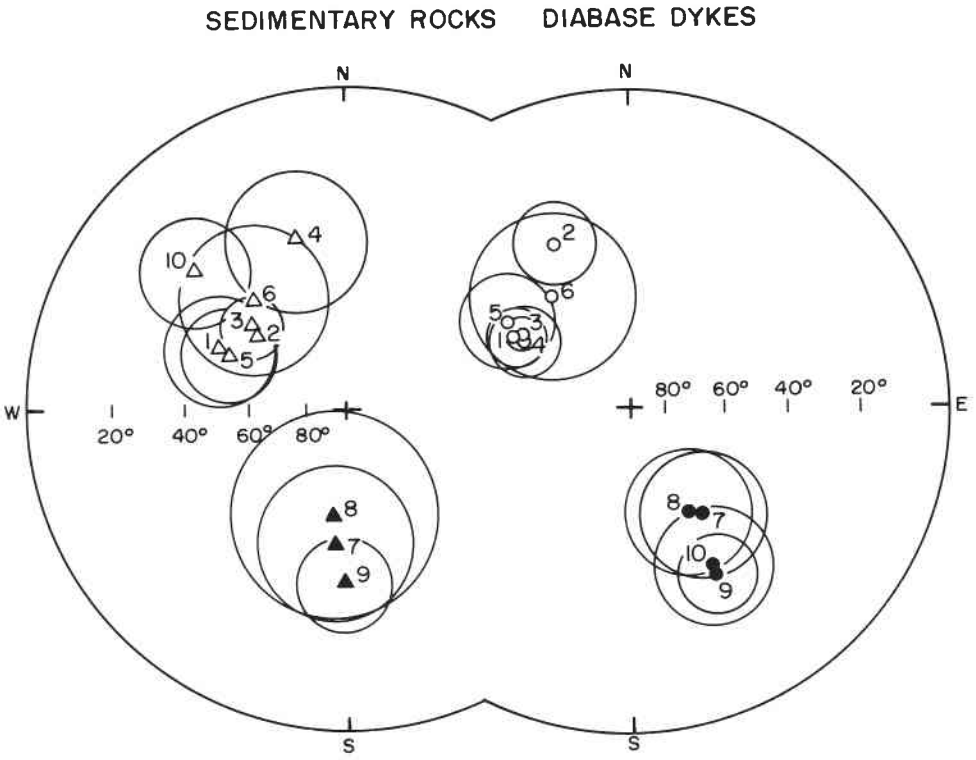


FIG. 7. Stereonet plot of the site-mean directions of remanence for the sedimentary rocks (left) and diabase dykes (right). The circular cone at the 95% confidence limit (Fisher statistics) is also indicated. The average cleaned magnetic vectors are significantly different for sites G-2, G-4, and G-10 and marginally different for sites G-7 and G-9. This approach brackets the age magnetization for components A and B.

rotation. (*iii*) A Kiaman overprint of Carbo-Permian age characterizes the main cluster of poles. In both DL and SP (Fig. 8), this overprint was detected in addition to the primary remanence. However, if the Kiaman overprint is so generalized in the Appalachians, why is it not also found in the present paleomagnetic study? Explanation *ii* is unlikely, but explanations *i* and *iii* are plausible; other reasons can certainly be considered. The important point to remember in this exercise is that the best known segment of the Paleozoic APWP of North America is still ambiguous.

The pole position of component A coincides better with the Late Devonian set of poles for North America (Seguin *et al.* 1982; Seguin 1986). The paleopole of component B is situated relatively close to paleopoles obtained on Middle Carboniferous red beds and volcanics (135°E, 21°N) for central New Brunswick (Seguin *et al.* 1985), from the Lower Carboniferous Deer Lake Group (128°E, 21°N) and Spout Falls Formation (140°E, 28°N) from western Newfoundland (Irving and Strong 1984; Murthy 1985), and the Lower Carboniferous Terrenceville Formation (124°E, 27°N) of eastern Newfoundland (Kent 1982). Averaging these paleopoles with the one of this study, an Early Carboniferous paleopole of North America is tentatively established with a mean at 138°E, 22°N. Independently of the reasons invoked to explain the divergence between the main cluster of Carboniferous paleomagnetic poles and the other set of poles to the southwest, this paleomagnetic investigation and a few other recent studies undertaken in the northern Appalachians suggest that little APWP motion ($\simeq 10°$) took place in the Late Devonian – Early Carboniferous time span, whereas a larger motion ($\simeq 20°$) occurred in the Early–Late Carboniferous interval.

Paleomagnetic data from the Carboniferous and Devonian lithological units of Quebec have important tectonic and magnetostratigraphic implications. These results demonstrate that Early–Middle Carboniferous time the paleopole position was the same through the whole section of the northern Appalachians, i.e., from the Avalon to the Humber zones (Williams 1979). Thus no lateral motion occurred later than at least Early Carboniferous time in this sector of the eastern margin of cratonic North America. Assuming that the time of intrusion of the diabase dykes occurred in the Early Carboniferous, the paleolatitude was then located around 25°S. This value is positioned to the north of the group of paleolatitudes (30–35°S) given by Irving and Strong (1985) for the Late Devonian aged rocks of southeastern Newfoundland.

The intrusion of the eastern Gaspé dykes probably took place at the same time as the plutons in north-central Gaspé (de Römer 1974) and some alkalic rocks in eastern and central United States (Zartman 1977). The age of the Gaspé intrusives ranges from Early Carboniferous to Late Devonian (315–360 Ma), while that of the same igneous rocks from eastern United States is about the same but has a more variable range. These geological correlations emphasize the postkinematic (very late to post-Acadian orogenic phase) character of these dykes (de Römer 1974; Zartman 1977). Assuming that the age of magnetization is correctly interpreted, this means that the compressive tectonic regime changed abruptly to a tensional regime at very Late Devonian time. As the dykes are oriented east–west indicating a north–south extension, the stresses at that time were orthogonal to the trend of the present North Atlantic Ocean. This situation suggests strongly that the Gaspé dykes are unrelated to the opening of the North Atlantic Ocean and that the very first continental extensions related to the actual North Atlantic Ocean did not begin until Middle to Late

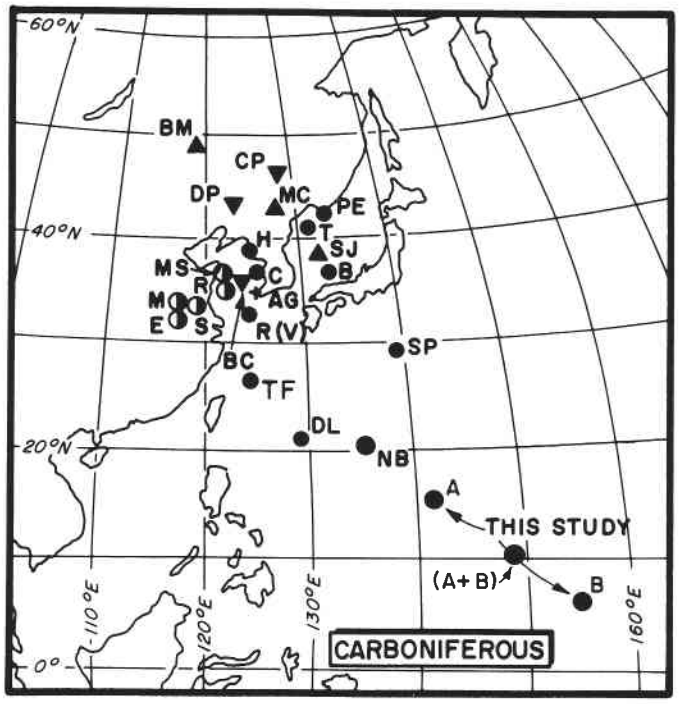


FIG. 8. Most-reliable paleopoles obtained from Carboniferous rock units from eastern North America. Modified from Roy *et al.* (1983). Additions: DL (Deer Lake Group—Irving and Strong 1984), NB (New Brunswick volcanics—Seguin *et al.* 1985), SP (Spout Falls Formation—Murthy 1985), and TF (Terrenceville Formation—Kent 1982).

Carboniferous time. In actual fact, the Late Triassic, Early Jurassic, and Early Cretaceous dykes from southern Quebec and central New England show mainly northeast trends (McHone 1978).

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BÉLAND, J. 1978. Les failles du bassin du Nord-Ouest et du Troisième Lac de l'est de la Gaspésie. Ministère de l'Énergie et des Ressources du Québec, DP-740.

BOURQUE, P. A. 1977. Le Silurien et le Dévonien basal du nord-est de la Gaspésie. Ministère des Richesses naturelles du Québec, ES-29.

BOURQUE, P. A., and LACHAMBRE, G. 1980. Stratigraphie du Silurien et du Dévonien basal du sud de la Gaspésie. Ministère de l'Énergie et des Ressources du Québec, ES-30.

BRISEBOIS, D. 1979. Géologie de la région de St-Georges-de-Malbaie. Ministère des Richesse naturelles du Québec, DPV-664.

CREER, K. M. 1968. Palaeozoic paleomagnetism. Nature (London), **219**: 246–250.

DERÖMER, H. S. 1974. Geology and age of some plutons in north–central Gaspé, Canada. Canadian Journal of Earth Sciences, **11**: 570–582.

DEUTSCH, E. R., and RAO, K. V. 1977. New palaeomagnetic evidence fails to support rotation of western Newfoundland, Nature (London), **266**: 314–317.

IRVING, E. 1979. Paleopoles and paleolatitudes of North America and speculation about displaced terrains. Canadian Journal of Earth Sciences, **16**: 669–694.

IRVING, E., and STRONG, D. F. 1984. Paleomagnetism of the Early Carboniferous Deer Lake Group, western Newfoundland: no evidence for Mid-Carboniferous displacement of Acadia. Earth and Planetary Science Letters, **691**: 379–391.

——— 1985. Paleomagnetism of rocks from Burin Peninsula, Newfoundland: hypothesis of late Paleozoic displacement of Acadia criticized. Journal of Geophysical Research, **90**: 1949–1962.

KENT, D. V. 1982. Paleomagnetic evidence for post-Devonian displacement of the Avalon Platform (Newfoundland). Journal of Geophysical Research, **87**: 8709–8716.

KENT, D. V., and OPDYKE, N. D. 1978. Paleomagnetism of the Devonian Catskill red beds: evidence for motion of the coastal New England – Canadian Maritime region relative to cratonic North America. Journal of Geophysical Research, **73**: 4441–4450.

KUMARAPELI, P. S., and SAULL, V. A. 1966. The St. Lawrence valley system: a North American equivalent of the East African Rift valley. Canadian Journal of Earth Sciences, **3**: 636–652.

MCGERRIGLE, H. W. 1950. La géologie de l'est de Gaspé. Ministère des Mines du Québec, RG-35.

MCHONE, J. G. 1978. Distribution, orientations, and ages of mafic dikes in central New England. Geological Society of American Bulletin, **89**: 1645–1655.

MURTHY, G. S. 1985. Paleomagnetism of certain constituents of the Bay St. George sub-basin, western Newfoundland. Physics of the Earth and Planetary Interiors, **39**: 89–107.

ROY, J. L., and MORRIS, W. A. 1983. A review of paleomagnetic results from the Carboniferous of North America: the concept of Carboniferous geomagnetic field horizon markers. Earth and Planetary Science Letters, **65**: 167–181.

ROY, J. L., TANCZYK, E., and LAPOINTE, P. 1983. The paleomagnetic record of the Appalachians. *In* Regional trends in the geology of the Appalachian–Caledonian–Hercynian–Mauritanide Orogen. *Edited by* P. E. Schenk. NATO ASI Series 116, pp. 11–26.

RUST, B. R. 1976. Stratigraphic relationships of the Malbaie Formation (Devonian), Gaspé, Quebec. Canadian Journal of Earth Sciences, **13**: 1556–1559.

SEGUIN, M. K. 1982. Geophysics of the Quebec Appalachians. Tectonophysics, **81**: 1–50.

——— 1985. Predlagaemaja Devonskaja traektorija paleomagnitnovo polioca cevernoie Ameriki. Geomagnetizm i Aèronomiya, **25**: 118–125.

——— 1986. Paleomagnetism of Lower Devonian units from Gaspé, Québec. Earth and Planetary Science Letters, **78**: 129–138.

SEGUIN, M. K., and GAHÉ, E. 1985. Paleomagnetism of Lower Devonian volcanics and Devonian dykes from northcentral New Brunswick, Canada. Physics of the Earth and Planetary Interiors, **38**: 262–276.

SEGUIN, M. K., and PETRYK, A. A. 1984. A paleomagnetic study of upper Orodovician – Lower Silurian strata of Anticosti Island, Quebec. Northeastern Geology, **6**: 25–37.

SEGUIN, M. K., RAO, K. V., VENUGOPAL, D. V., and GAHÉ, E. 1981. Paleomagnetism of parts of the Late Triassic diabase dike system associated with the trans-New Brunswick aeromagnetic lineament. Canadian Journal of Earth Sciences, **18**: 1776–1787.

SEGUIN, M. K., RAO, K. V., and PINEAULT, R. 1982. Paleomagnetic study of Devonian rocks from Ste-Cécile–St-Sébastien region, Quebec Appalachians. Journal of Geophysical Research, **87**: 7853–7864.

SEGUIN, M. K., SINGH, A., and FYFFE, L. 1985. New paleomagnetic data from Carboniferous volcanics and red beds from central New Brunswick. Geophysical Research Letters, **12**: 81–84.

SIKANDER, A. H. 1978. The Gaspé Sandstone, eastern Gaspé. Ministère des Richesses naturelles du Québec, GM-33690.

SPARIOSU, D. J., and KENT, D. V. 1983. Paleomagnetism of the

Lower Devonian Traveler Felsite and the Acadian orogeny in the New England Appalachians. Geological Society of America Bulletin, **94**: 1319–1328.

STEVENS, A. E. 1980. Reexamination of some larger La Malbaie, Quebec, earthquakes (1924–1978). Bulletin of the Seismological Society of America, **70**: 529–557.

VAN DER VOO, R. 1981. Paleomagnetism of North America: a brief review. *In* Paleoreconstruction of the continents. *Edited by* M. W. McElhinny and D. A. Valencio. American Geophysical Union and Geological Society of America, Geodynamics Series 2, pp. 159–176.

WANLESS, R. K., STEVENS, R. D., LACHANCE, G. R., and RIMSAÏTE, J. Y. H. 1966. Age determinations and geological studies: K–Ar isotopic ages, report 7. Geological Survey of Canada, Paper 66-17, pp. 98–100.

WANLESS, R. K., STEVENS, R. D., LACHANCE, G. R., and DELABIO, R. N. 1973. Age determinations and geological studies: K–Ar isotopic ages, report 11. Geological Survey of Canada, Paper 73-2. pp. 72–74.

WILLIAMS, H. 1979. Appalachian Orogeny in Canada. Canadian Journal of Earth Sciences, **16**: 792–807.

ZARTMAN, R. E. 1977. Geochonology of some alkalic rock provinces in eastern and central United States. Annual Review of Earth and Planetary Sciences, **5**: 257–286.